

Campus Energy Saving System

Common Troubleshooting Scenarios – Trainer/Assignment Worksheet

No.	Problem / Scenario	Possible Cause	Troubleshooting / Solution	Expected Result
1	Lights remain ON when the room is empty	Motion sensor is not detecting occupancy or automation is disabled.	Check sensor position, power supply and automation settings. Test the sensor manually.	Lights turn OFF automatically when the room is unoccupied.
2	Lights do not turn ON automatically	Faulty motion sensor, disconnected wiring or low sensor power.	Check sensor connection and power. Clean/reposition the sensor and test detection.	Lights activate when someone enters the room.
3	AC remains ON after students leave	Occupancy sensor or scheduled control is not working.	Verify occupancy detection and AC schedule. Check whether manual override is enabled.	AC switches OFF or enters energy-saving mode in an empty room.
4	Energy consumption suddenly increases	Equipment left running, faulty appliance or abnormal usage.	Check the energy dashboard and identify the area with unusual consumption. Inspect equipment.	Unnecessary energy use is identified and reduced.
5	Energy meter shows incorrect readings	Meter communication, sensor or calibration issue.	Check meter connections, communication status and calibration. Restart monitoring device if needed.	Accurate energy readings are displayed.
6	Dashboard is not updating	Network connection failure or sensor communication problem.	Check network connectivity, gateway status and sensor connection. Restart gateway if necessary.	Dashboard displays current energy data.
7	One building has unusually high consumption	High equipment usage, faulty equipment, or lights/AC left ON.	Compare buildings on the dashboard and conduct a physical inspection of the high-use area.	High-consumption areas are identified and corrective action is taken.
8	Energy-saving alert is not received	Notification settings disabled or threshold incorrectly configured.	Check notification permissions and alert thresholds. Enable required notifications.	Users receive alerts for abnormal consumption.
9	Students ignore energy-saving alerts	Alerts are too frequent or not useful/engaging.	Make alerts short and actionable. Add awareness messages, points or rewards.	Student participation and awareness improve.
10	Solar power system produces less energy	Panel shading, dust, weather conditions or system fault.	Inspect panels, remove dust safely, check shading and inspect inverter/system status.	Solar generation improves or the fault is identified.
11	Smart switch is not responding	Device disconnected, power failure or communication error.	Check power and network connection. Reconnect or reset the smart switch according to procedure.	Device responds to commands normally.
12	Equipment switches OFF at the wrong time	Incorrect schedule or system clock configuration.	Verify system clock, timetable, holidays and automation rules.	Equipment operates according to the correct campus schedule.
13	System works in one classroom but not another	Different sensor configuration or device failure.	Compare settings and hardware. Test the affected device individually.	Both classrooms operate correctly.
14	Students report lights are too dim	Incorrect brightness setting or faulty lighting equipment.	Check lighting levels and dimmer settings. Replace faulty lamps if required.	Comfortable lighting is maintained without unnecessary energy use.
15	System stops after a power outage	Controller did not restart correctly or configuration was lost.	Check controller/gateway power, restart system and verify configuration and device connections.	Energy management system resumes normal operation.

Basic Troubleshooting Procedure

1. Identify the problem → 2. Check power supply → 3. Check sensors/devices → 4. Check network connection → 5. Check software/settings → 6. Test the system → 7. Record the solution

Example Scenario: Classroom lights remain ON after students leave

Investigation: Check motion-sensor power and detection, verify automatic lighting settings, check manual override, and test the system with an empty classroom. Solution: Reposition/reconnect the sensor and restore the automatic OFF setting.
Expected outcome: Lights switch OFF after the configured unoccupied period, reducing unnecessary electricity consumption.